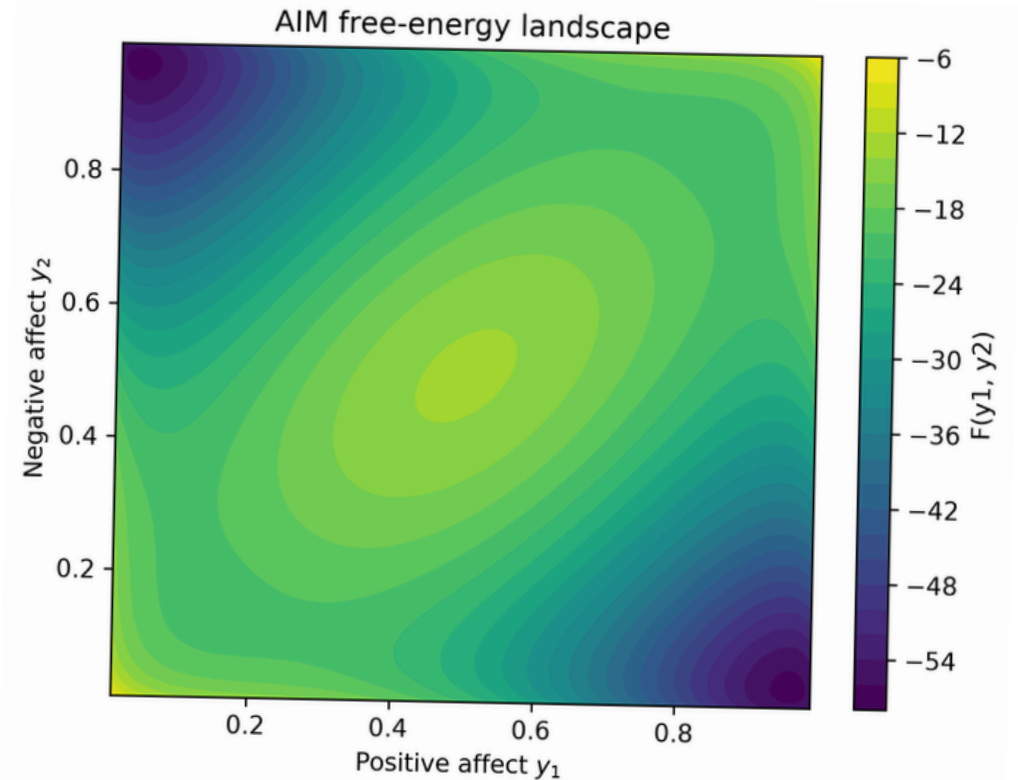


When does a temporary negative event change an affective state?

Reproducing and extending the Affective Ising Model with temporal negative inputs

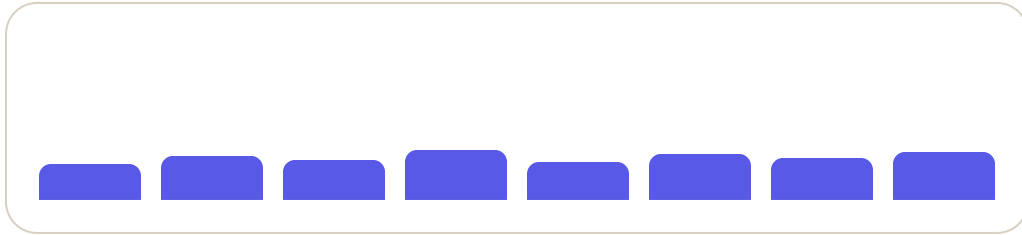
Rosa Graells Danús



The same average affect can hide radically different dynamics

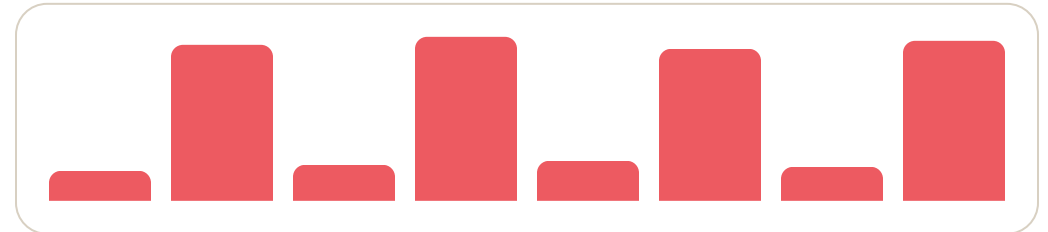
Person A

Usually near the same emotional state, with small fluctuations.



Person B

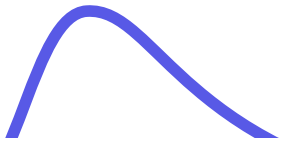
Alternates between two recurring states, but can have the same mean.



The paper's question: can one stochastic model explain skewness, changing variance and multimodality—not only the mean?

Affective data are often non-Gaussian

Skewness



Mass accumulates near one boundary.

Changing variance



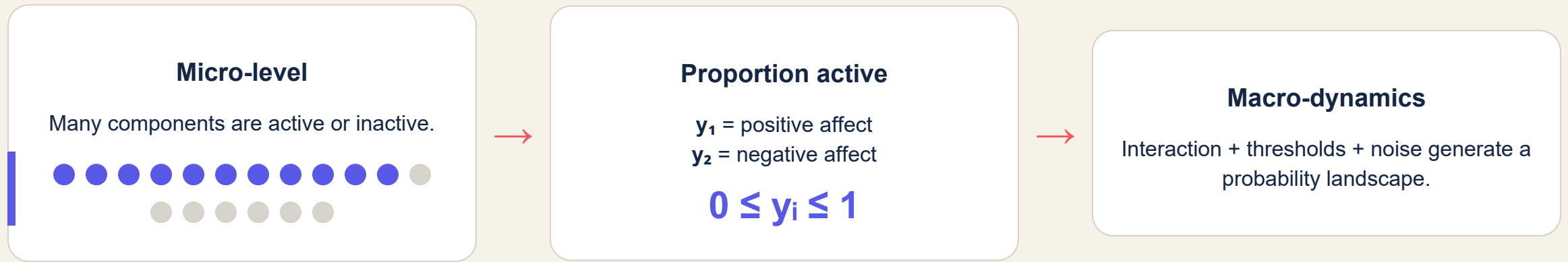
Fluctuation size differs across states.

Multimodality



Two affective “homebases” can coexist.

Many binary micro-components create a continuous affective state



The model is not a mood label. It is a stochastic system whose states are **proportions of active affective components**.

Before seeing the equations: where will the system spend most of its time?

Rule 1 · self-reinforcement

Active components make similar components easier to activate.

clusters form

Rule 2 · cross-inhibition

Positive and negative components oppose simultaneous activation.

mixed states become costly

PA basin

high y_1 · low y_2

ridge

unstable mixed region

NA basin

low y_1 · high y_2

Four parameter families shape the affective landscape

Λ_i

within-valence coupling

Θ_i

activation threshold

Λ_{12}

cross-valence inhibition

N_i

number of components

Energy

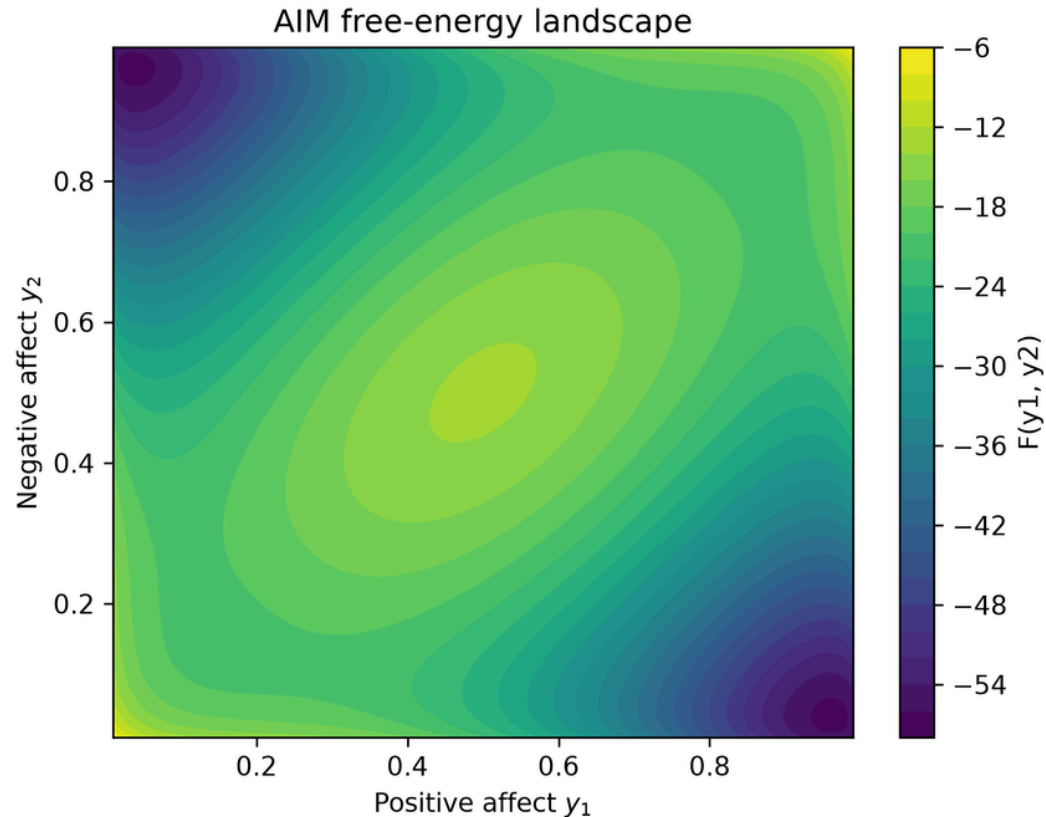
Encodes which configurations are easier or harder.

Entropy

Counts how many micro-configurations correspond to each proportion.

Free energy = interaction/threshold terms + combinatorial entropy. This is the bridge from Ising components to a continuous stochastic model.

Homebases are minima of free energy



- 1 Low F means a more probable state.
- 2 Two minima correspond to PA- and NA-dominated basins.
- 3 The ridge is the barrier between them.
- 4 Noise can still produce rare barrier crossings.

A temporary input can matter after it ends if it pushes the trajectory across this ridge.

The Boltzmann distribution converts free energy into probability

$$p(y_1, y_2) = \frac{\exp[-\beta F(y_1, y_2)]}{Z}, \quad Z = \int_0^1 \int_0^1 \exp[-\beta F(y_1, y_2)] dy_1 dy_2.$$

 β

inverse stochastic temperature

 $F(y_1, y_2)$

free-energy landscape

 Z

normalization constant

As βF increases, probability decreases exponentially. Local minima of F become modes of p .

Interaction, thresholds and entropy appear in one potential

$$F(y_1, y_2) = \sum_{i=1}^2 (-\Lambda_i y_i^2 + \Theta_i y_i) + \Lambda_{12} y_1 y_2 + \sum_{i=1}^2 \frac{N_i}{\beta} [y_i \ln y_i + (1 - y_i) \ln(1 - y_i)] .$$

Energetic contribution

$$-\Lambda_i y_i^2 + \Theta_i y_i + \Lambda_{12} y_1 y_2$$

Combinatorial contribution

$$(N_i/\beta)[y_i \ln y_i + (1-y_i)\ln(1-y_i)]$$

$$\frac{\partial F}{\partial y_i} = -2\Lambda_i y_i + \Theta_i + \Lambda_{12} y_j + \frac{N_i}{\beta} \ln \left(\frac{y_i}{1 - y_i} \right), \quad j \neq i.$$

A Langevin SDE combines downhill drift and random fluctuations

$$dy_i(t) = -\beta D \frac{\partial F(\mathbf{y}(t))}{\partial y_i} dt + \sqrt{2D} dW_i(t), \quad i = 1, 2.$$

Drift

$-\beta D \nabla F$ pulls the state downhill.

Noise

$\sqrt{(2D)}dW$ allows stochastic exploration.

Pace D

Changes time scale, not the equilibrium shape.

The same model explains both local fluctuations and rare state changes

Inside a basin

Noise produces small, mean-reverting fluctuations.



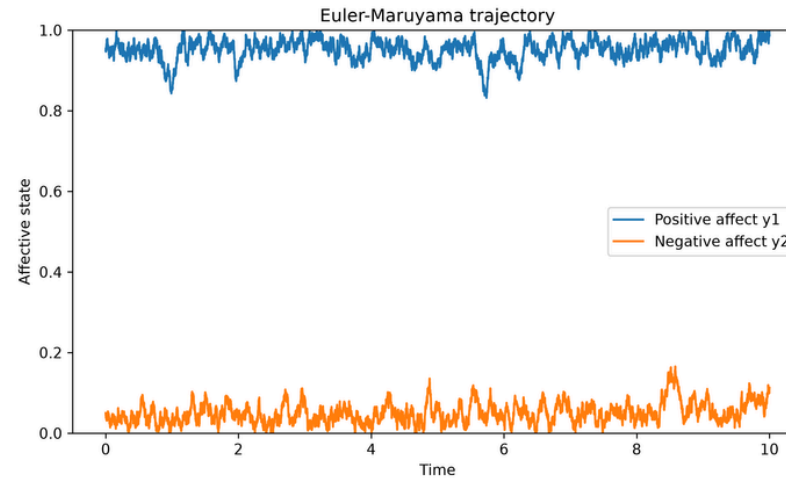
Near the ridge

Drift is weaker and perturbations become consequential.



Across the barrier

Drift now stabilizes the alternative basin.

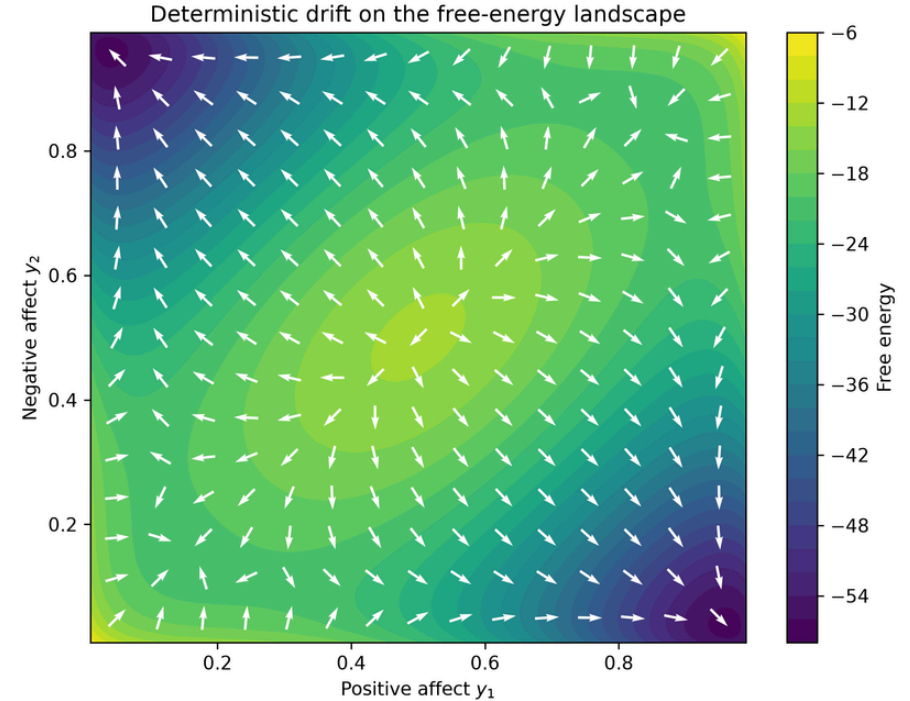
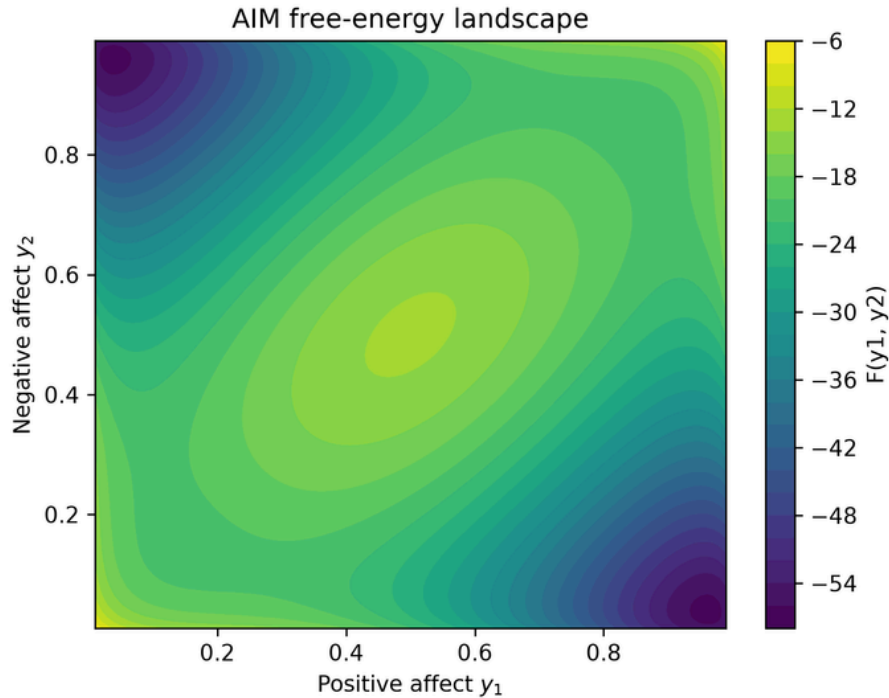


Reproduce first; extend only after the numerical mechanism is validated



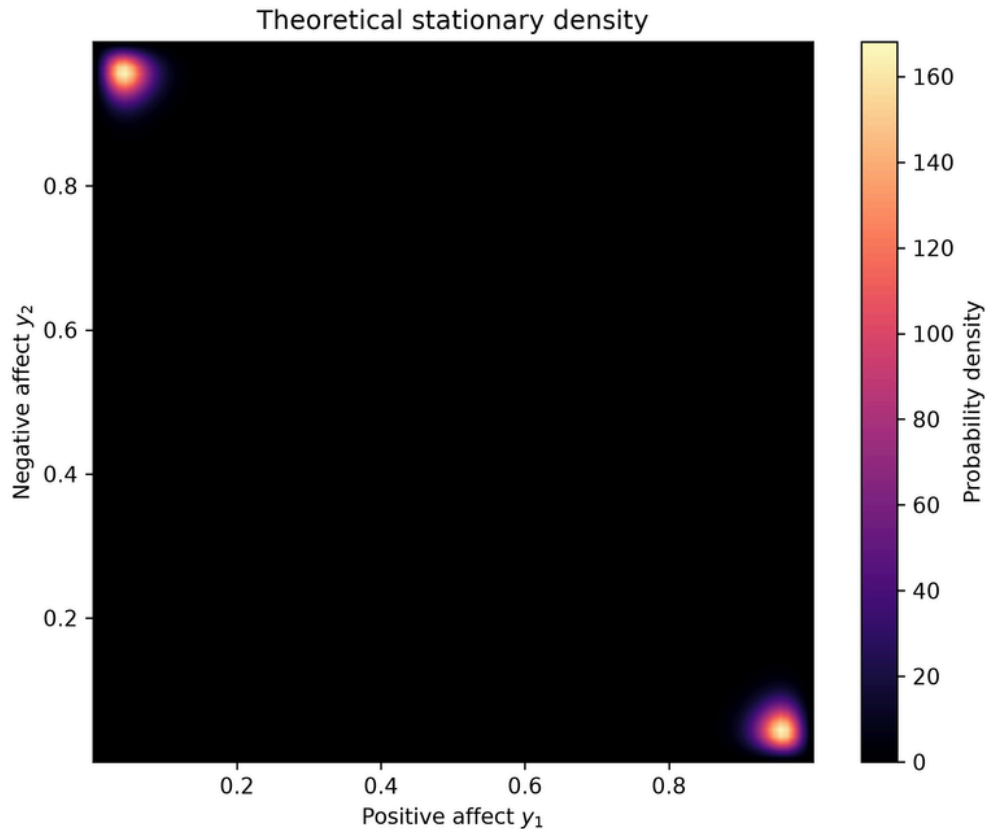
Research question: if total negative input is held fixed, does its temporal organization alter the probability of switching basins?

The implementation produces two basins and downhill drift



Gradient check passed: analytical and finite-difference derivatives agree to approximately 10^{-9} .

The landscape predicts a two-peak stationary density

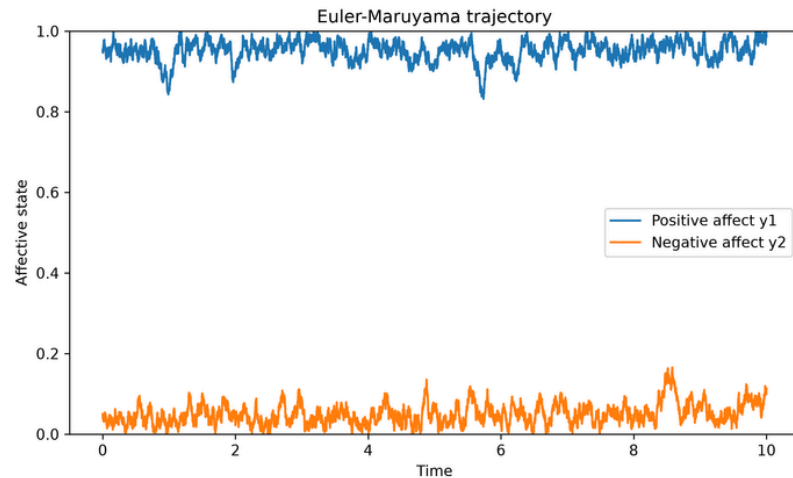


- 1 Evaluate F on $[\varepsilon, 1-\varepsilon]^2$.
- 2 Compute $\exp[-\beta(F-F_{\min})]$.
- 3 Integrate numerically to obtain Z .
- 4 Verify $\iint p(y_1, y_2) dy_1 dy_2 \approx 1$.

The two probability peaks coincide with the two free-energy minima.

Euler–Maruyama turns the SDE into reproducible trajectories

$$\mathbf{y}_{n+1} = \Pi_{[\varepsilon, 1-\varepsilon]^2} \left[\mathbf{y}_n - \beta D \nabla F(\mathbf{y}_n) \Delta t + \sqrt{2D \Delta t} \boldsymbol{\xi}_n \right], \quad \boldsymbol{\xi}_n \sim \mathcal{N}(\mathbf{0}, I_2).$$



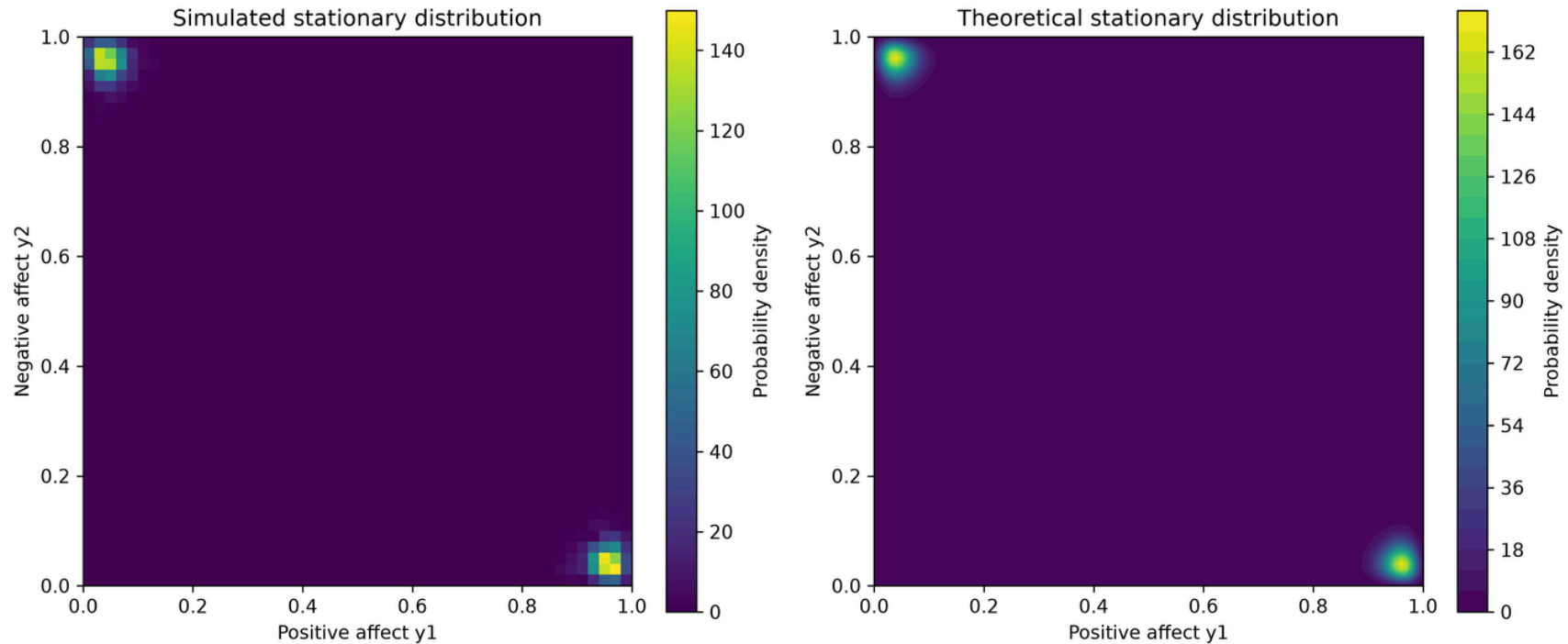
drift scales as Δt

noise scales as $\sqrt{\Delta t}$

seed makes a realization reproducible

clipping avoids log singularities

Simulation reproduces the theoretical stationary structure



Basin	Empirical mode	Theoretical mode	Error
PA	(0.97, 0.03)	(0.9596, 0.0404)	0.0147
NA	(0.03, 0.97)	(0.0404, 0.9596)	0.0147

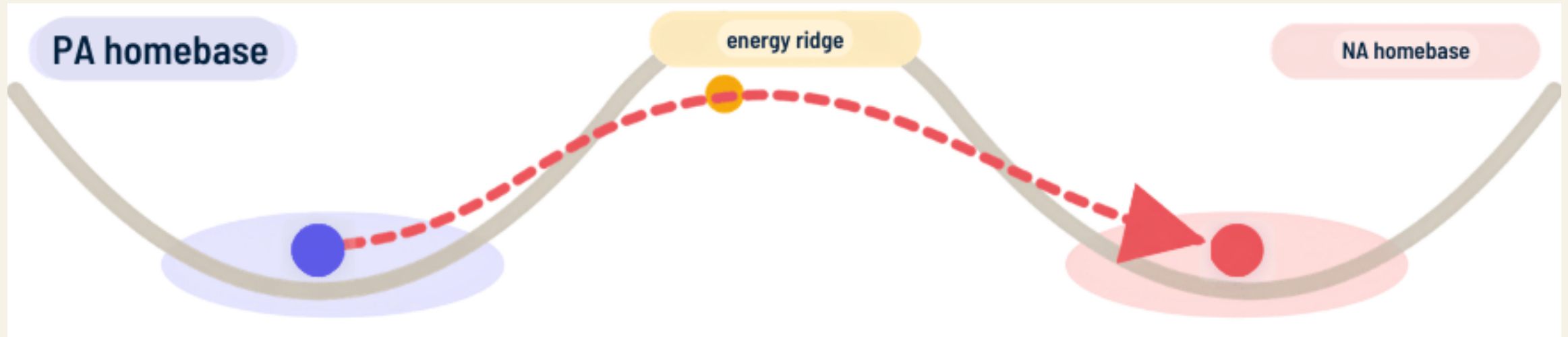
From a temporary event to a persistent state change

1 · BEFORE

The trajectory starts near the PA homebase: $y_0 = (0.96, 0.04)$.

2 · DURING THE EVENT

$B_2(t)$ tilts the landscape and pushes the trajectory toward the energy ridge.



3 · AFTER IT ENDS · If $y_2(T) > y_1(T)$, the trajectory is in the NA basin. The input is temporary; the switch can persist.

INPUT MODEL

How does a negative pulse enter the model?

TEMPORAL INPUT

$$B2(t) = A \leftarrow [t_o, t_o+t)$$

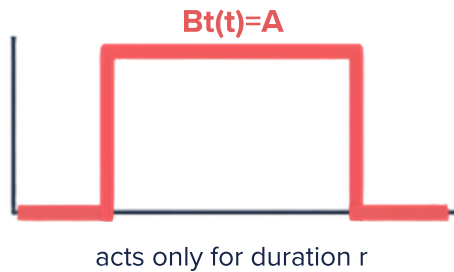
$$\theta_2 \rightarrow \theta_2 - B2(t)$$

THREE CONTROLS

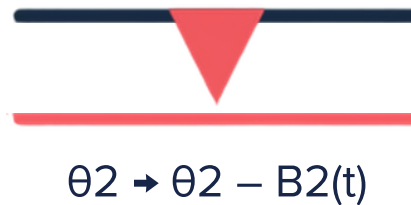
A intensity **t** onset **T** duration

MECHANISM — READ FROM LEFT TO RIGHT

1 • Finite pulse



2 • Threshold decreases



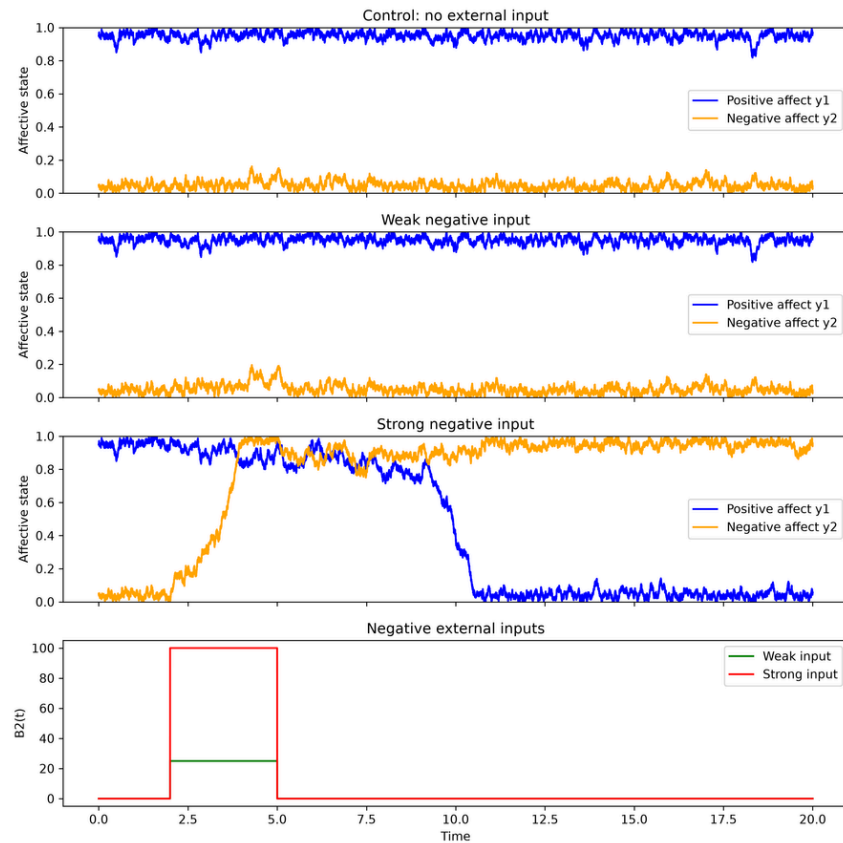
3 • Drift toward higher y



SCIENTIFIC INTERPRETATION

$Bs(t) > 0$ lowers $\theta_2 \rightarrow$ increases the y drift — raises the probability of crossing into the NA basin.

With the same noise, weak input perturbs and strong input switches



Control

 $A=0$

Weak

 $A=25 \cdot \tau=3$

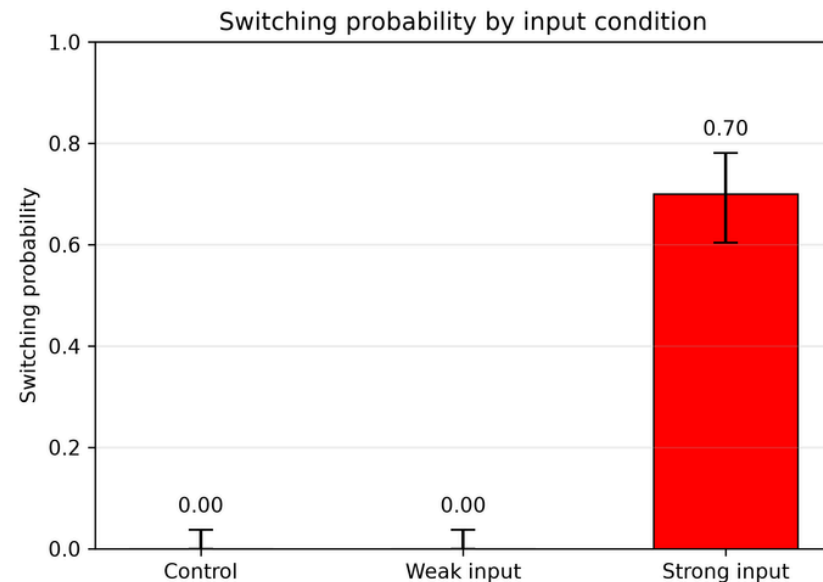
Strong

 $A=100 \cdot \tau=3$

Same seed isolates the input effect in this illustration.

How often does a trajectory finish in the negative basin?

$$m(t) = y_1(t) - y_2(t), \quad \text{switch at } T \iff m(T) < 0, \quad t_{\text{rec}} = \inf\{t \geq t_{\text{off}} : m(s) > 0 \forall s \in [t, t + w]\} - t_{\text{off}}.$$

**0/100**

control

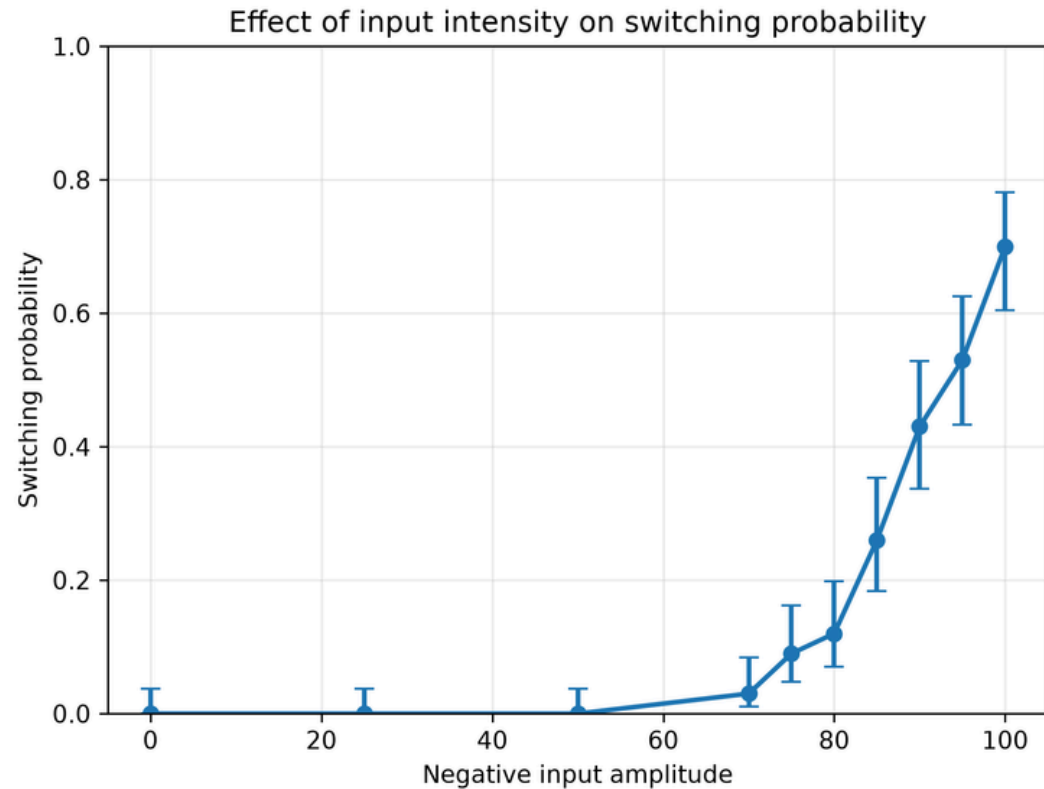
0/100

weak

70/100

strong

Switching rises nonlinearly once input approaches the barrier



0–50

no observed switches

70–80

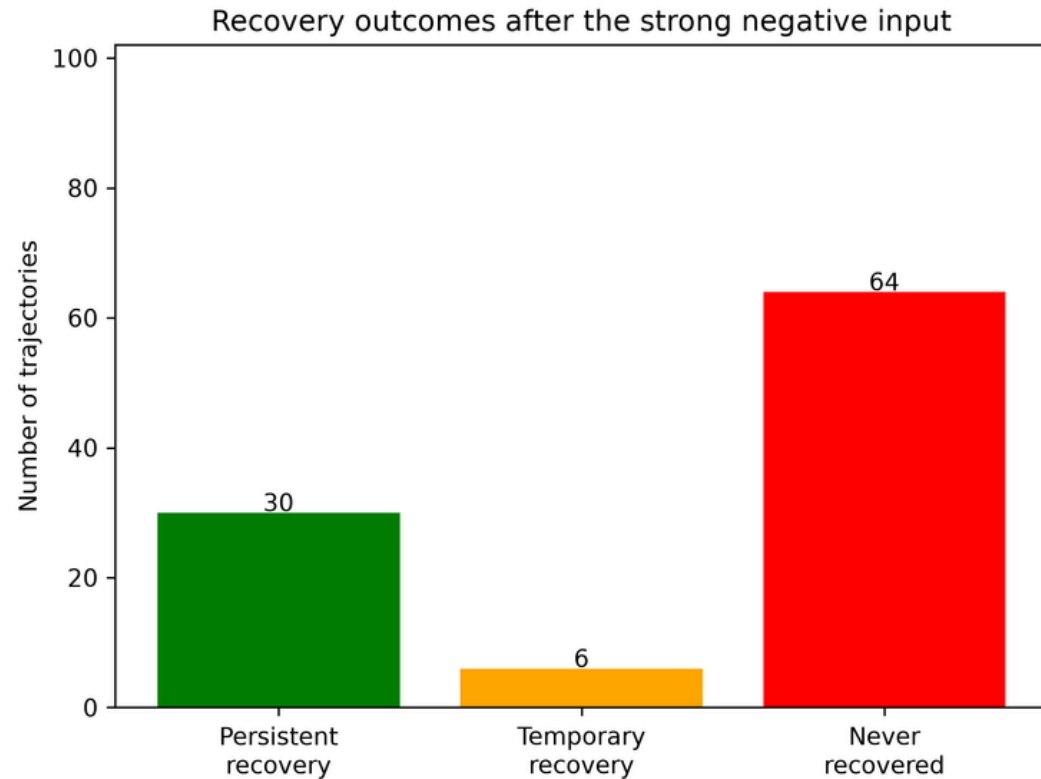
rare switches begin

85–100

rapid increase to 0.70

Near the barrier, small amplitude changes alter outcome probability strongly.

Recovery requires a stable return—not one noisy crossing



30

persistent recovery

6

temporary recovery

64

never recovered

Mean observed recovery time: **1.35 abstract units** among 36 recovered paths.

Stable-window criterion: $w=0.5$ abstract units (1000 integration steps).

Same total input, three different temporal shapes



Acute

One intense argument: high amplitude concentrated in a short interval.

$$A=100 \cdot \tau=3 \cdot \text{area}=300$$



Prolonged

Steady rain all week: lower amplitude sustained for longer.

$$A=50 \cdot \tau=6 \cdot \text{area}=300$$



Fragmented

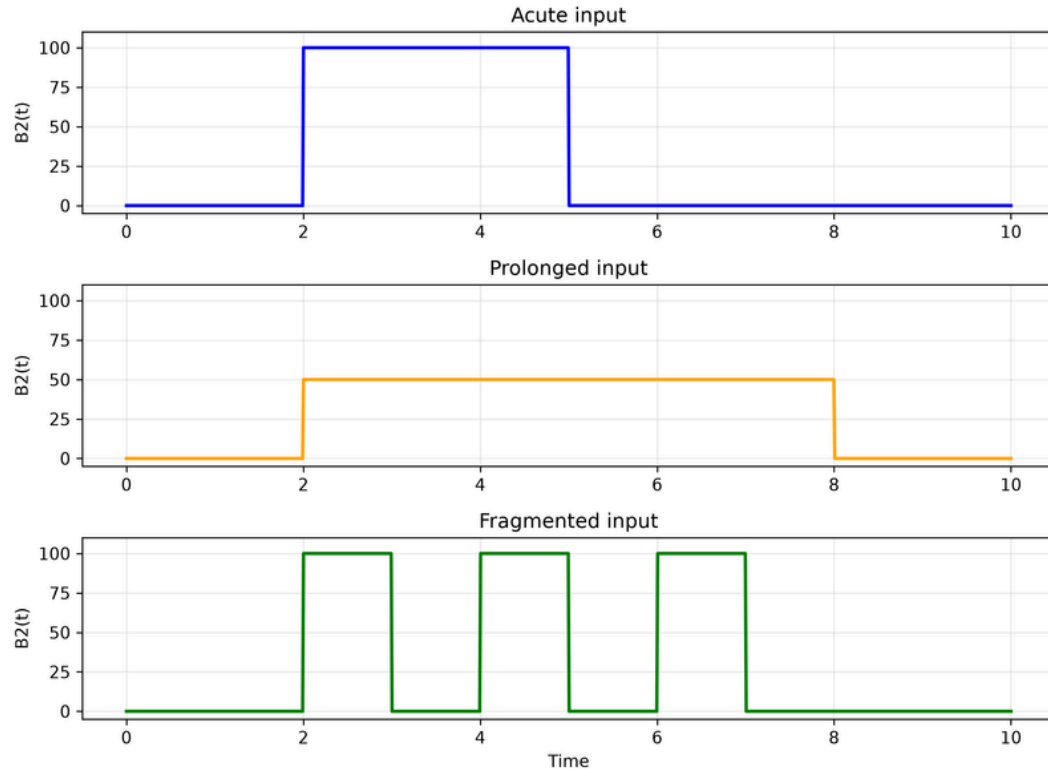
Three stressful messages separated by pauses.

$$3 \text{ pulses} \cdot A=100 \cdot \tau=1 \cdot \text{area}=300$$

The examples are analogies for timing. Scientifically, $\int B_2(t)dt=300$ is held constant and only temporal shape changes.

Equal-area inputs isolate temporal organization

Negative inputs with equal total area

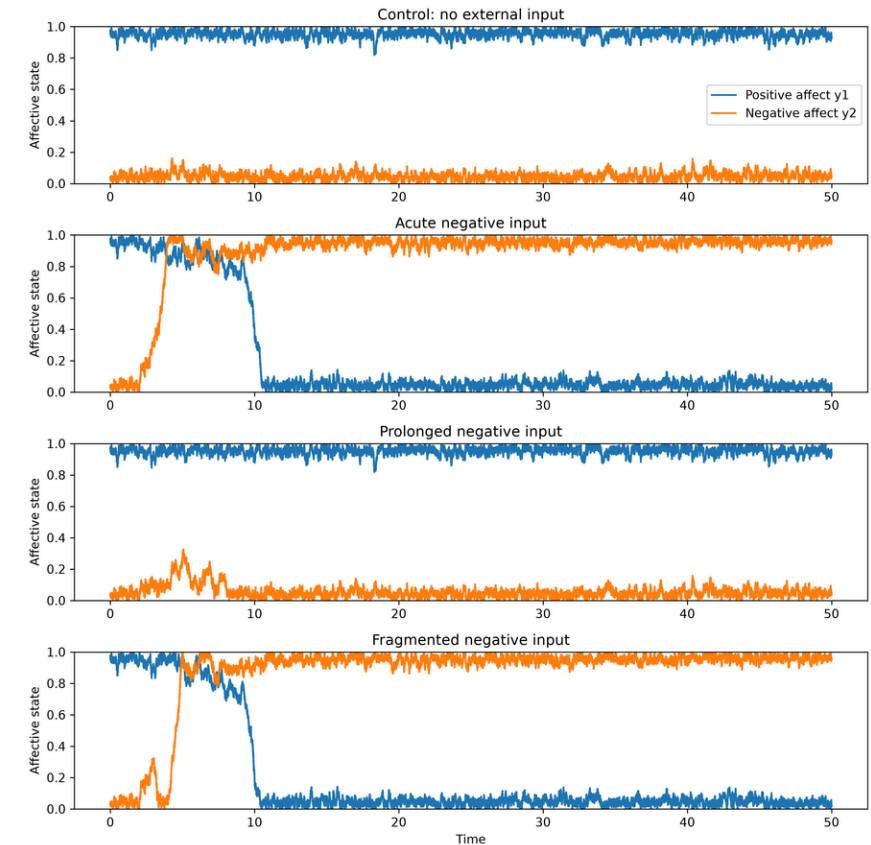
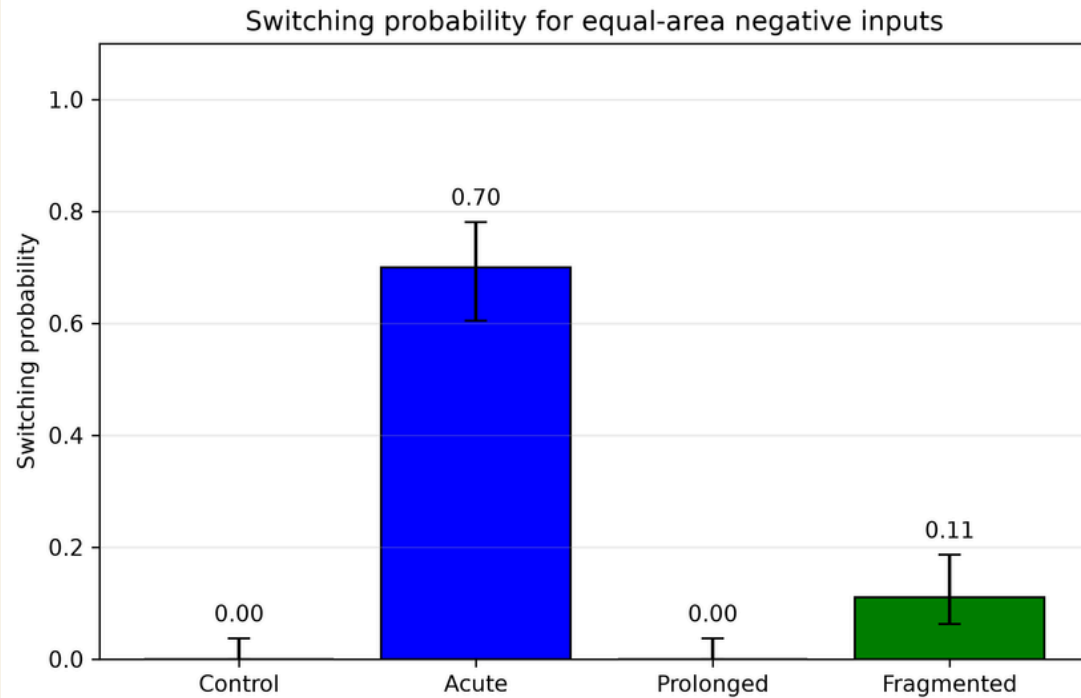


$$\int B_2(t) dt = 300, \quad 100 \times 3 = 50 \times 6 = 3 \times 100 \times 1.$$

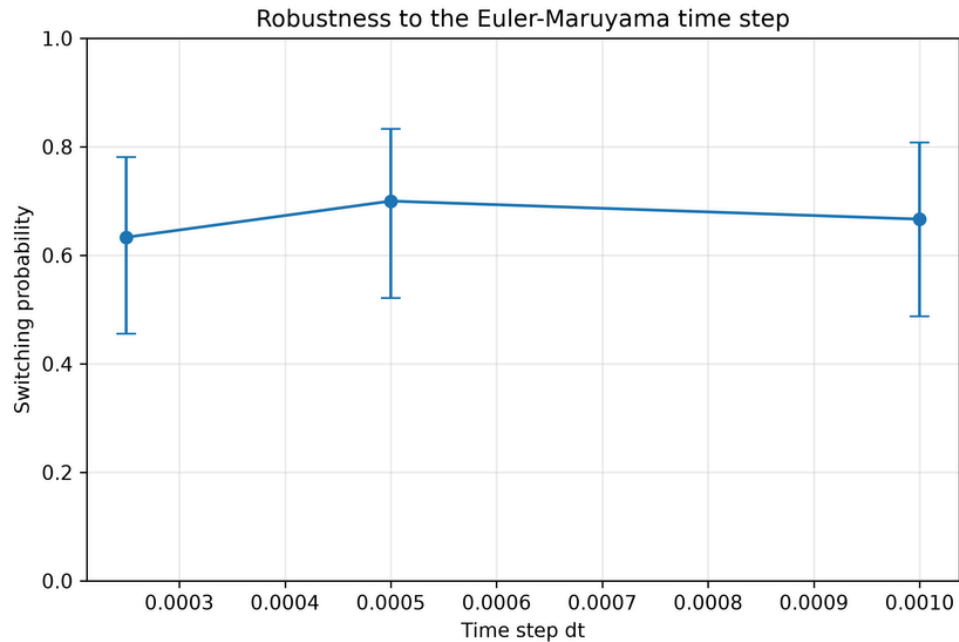
Input	A	Timing	Area
Acute	100	$\tau=3$	300
Prolonged	50	$\tau=6$	300
Fragmented	100	$3 \times \tau=1$	300

Amplitude, duration and spacing change; total accumulated input does not.

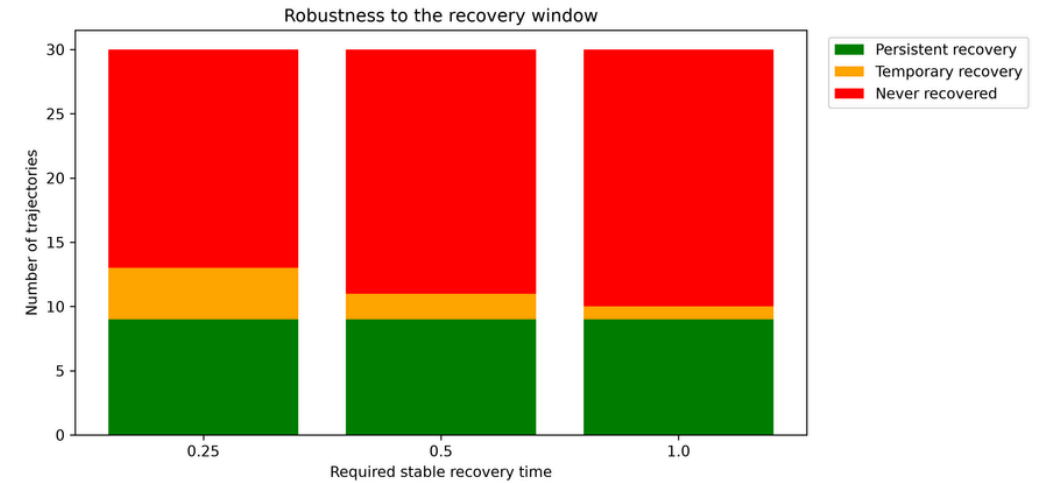
Equal accumulated input does not produce equal switching



The central result is stable to the tested numerical choices



Δt : .001 $\rightarrow$.667 $\cdot$.0005 $\rightarrow$.700 $\cdot$.00025 $\rightarrow$.633



Persistent count remains 9/30 for every tested window.

Numerical robustness supports the computation; it does not establish external or clinical validity.

CONCLUSION

Temporal structure changes the outcome



Main conclusion

A temporary negative input can produce a persistent basin switch.
Equal total input does not imply equal outcomes: timing and duration matter.



Computational and mechanistic result—not a clinical prediction.



Thank you